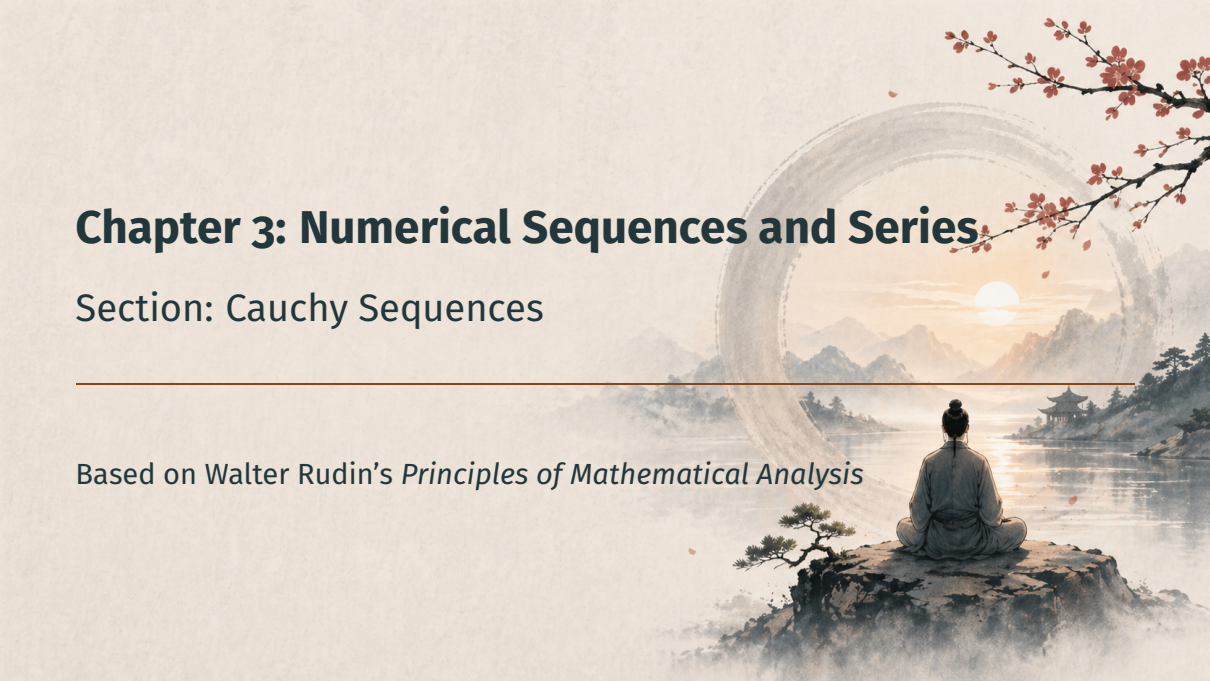


Chapter 3: Numerical Sequences and Series

The background is a traditional Chinese ink wash painting. A person with their hair in a bun, wearing a grey robe, sits in a meditative lotus position on a dark, craggy rock in the foreground. They are looking out over a calm lake that reflects the sky. In the distance, misty mountains rise. A large, faint, circular frame, resembling a ring or a portal, is centered in the background, framing the sun. A small pavilion with a traditional roof is visible on the right side of the lake. Red cherry blossom branches with small flowers hang down from the top right corner. A few red petals are scattered in the air and on the water.

Section: Cauchy Sequences

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

Cauchy Sequences and Diameter

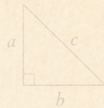
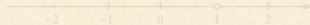
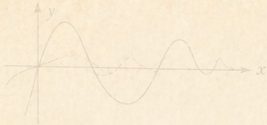
Diameters and Nested Compacts

The Cauchy Criterion

Complete Metric Spaces

Monotonic Sequences in $\mathbb{R}$

Summary



$$a^2 + b^2 = c^2$$

Motivation



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Why Cauchy Sequences?

- Convergence as we defined it **names a limit** in advance



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

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Why Cauchy Sequences?

- Convergence as we defined it **names a limit** in advance
- Often we want to know convergence *without* knowing the limit



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Why Cauchy Sequences?

- Convergence as we defined it **names a limit** in advance
- Often we want to know convergence *without* knowing the limit
- **Cauchy**: the terms eventually crowd together among themselves



$$f(x) = \sin x$$



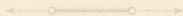
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Why Cauchy Sequences?

- Convergence as we defined it **names a limit** in advance
- Often we want to know convergence *without* knowing the limit
- **Cauchy**: the terms eventually crowd together among themselves
- In nice spaces (compact, $\mathbb{R}^k$): **Cauchy** $\iff$ **convergent**

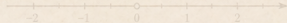


$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Cauchy Sequences and Diameter



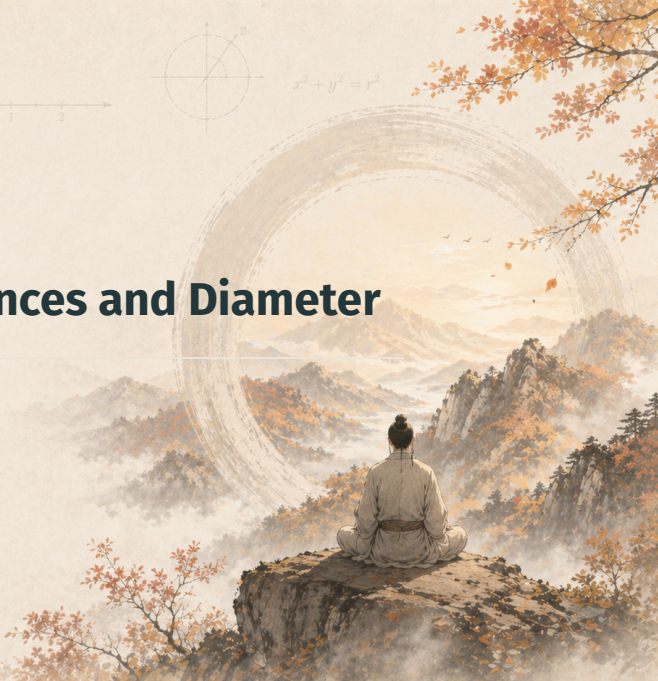
$$x^2 + y^2 = r^2$$



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$



Definition 3.8: Cauchy Sequence

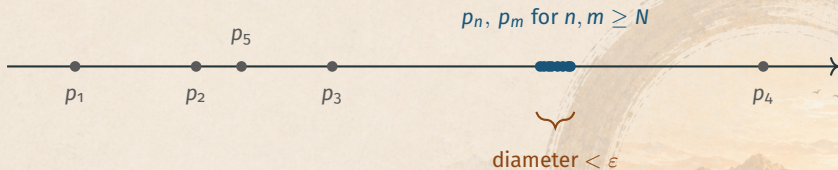
Definition 3.8 — Cauchy Sequence

A sequence $\{p_n\}$ in a metric space X is a **Cauchy sequence** if

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N} : \quad d(p_n, p_m) < \varepsilon \quad \text{whenever } n, m \geq N.$$

Key idea: Past index N , any two terms are within ε of each other.

Picture: Cauchy Means the Tail Is Tiny



Past index N , all terms fit inside an ε -window.

Definition 3.9: Diameter

Definition 3.9 — Diameter

Let E be a nonempty subset of a metric space X , and let

$$S = \{ d(p, q) : p \in E, q \in E \}.$$

The **diameter** of E is

$$\text{diam } E = \sup S.$$

Intuition: $\text{diam } E$ is the *largest distance* between two points of E .

Cauchy in Terms of Diameter

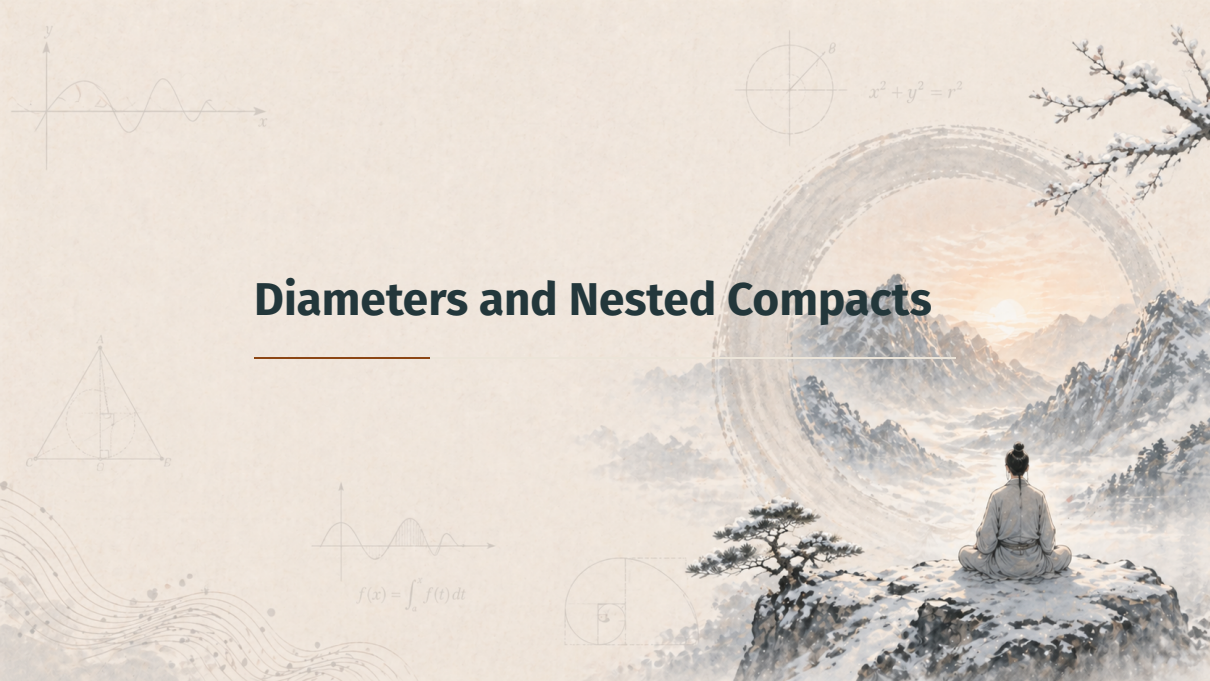
Let $E_N = \{p_N, p_{N+1}, p_{N+2}, \dots\}$ be the *tail* starting at index N .

Equivalent Form of the Cauchy Condition

$$\{p_n\} \text{ is Cauchy} \iff \lim_{N \rightarrow \infty} \text{diam } E_N = 0.$$

Why: $\text{diam } E_N < \varepsilon$ says that any two terms past index N are within ε .

Diameters and Nested Compacts

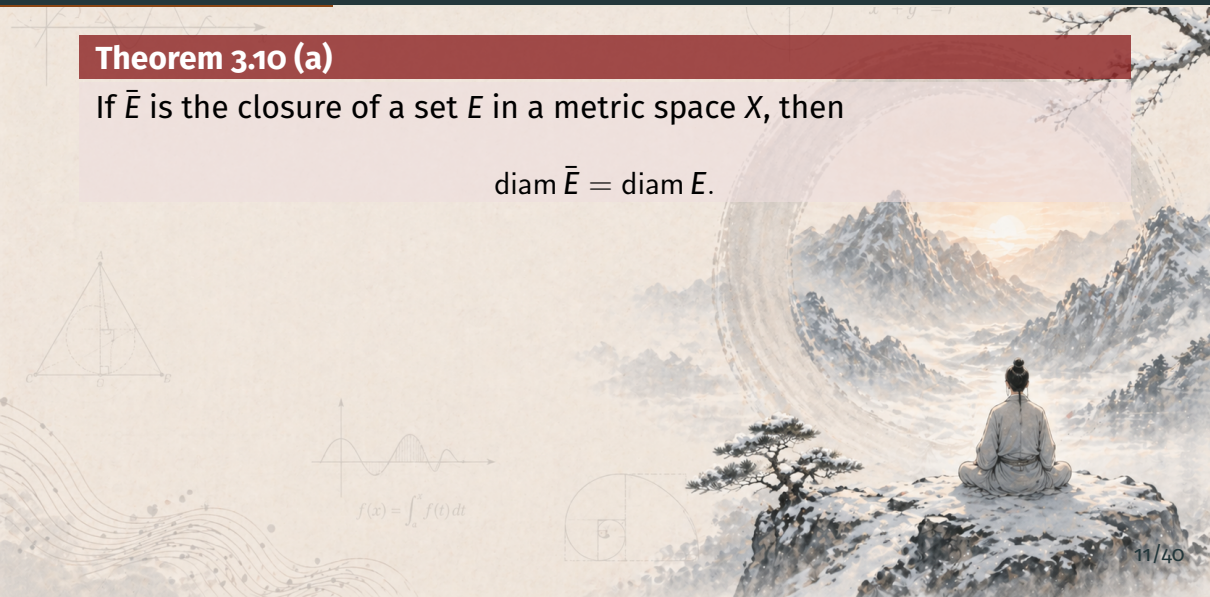


Theorem 3.10: Two Geometric Facts

Theorem 3.10 (a)

If $\bar{E}$ is the closure of a set E in a metric space X , then

$$\text{diam } \bar{E} = \text{diam } E.$$



Theorem 3.10: Two Geometric Facts

Theorem 3.10 (a)

If $\bar{E}$ is the closure of a set E in a metric space X , then

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Theorem 3.10 (b)

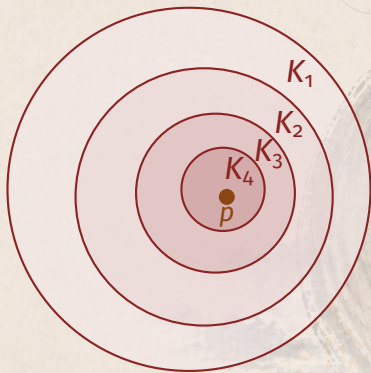
Let $\{K_n\}$ be a sequence of **nonempty compact** sets in X with

$$K_1 \supset K_2 \supset K_3 \supset \cdots \quad \text{and} \quad \lim_{n \rightarrow \infty} \text{diam } K_n = 0.$$

Then $\bigcap_{n=1}^{\infty} K_n$ consists of **exactly one point**.

$$f(x) = \int_a^x f(t) dt$$

Picture: Nested Compacts Pinch to a Point



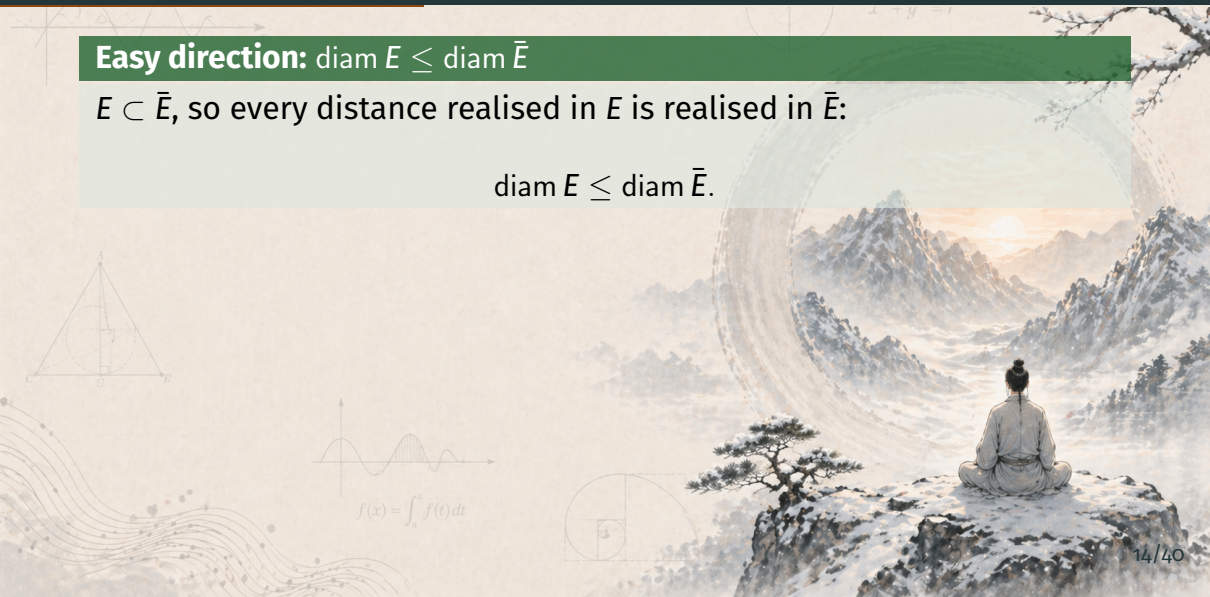
Each K_n is nonempty and compact; $K_{n+1} \subset K_n$; diameters shrink to 0; the intersection is a single point.

Proof of 3.10 (a): $\text{diam } \bar{E} = \text{diam } E$

Easy direction: $\text{diam } E \leq \text{diam } \bar{E}$

$E \subset \bar{E}$, so every distance realised in E is realised in $\bar{E}$:

$$\text{diam } E \leq \text{diam } \bar{E}.$$



Proof of 3.10 (a): $\text{diam } \bar{E} = \text{diam } E$

Easy direction: $\text{diam } E \leq \text{diam } \bar{E}$

$E \subset \bar{E}$, so $\text{diam } E \leq \text{diam } \bar{E}$.

Other direction: $\text{diam } \bar{E} \leq \text{diam } E$

Fix $\varepsilon > 0$, $p, q \in \bar{E}$. Pick $p', q' \in E$ with $d(p, p') < \varepsilon$ and $d(q, q') < \varepsilon$. Then

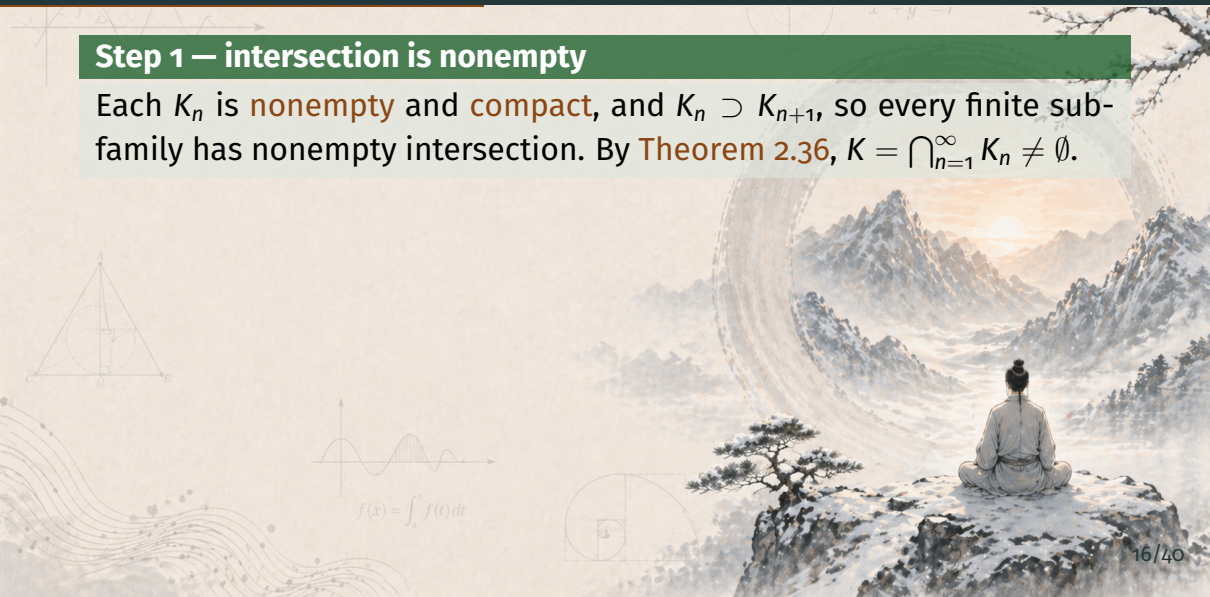
$$d(p, q) \leq d(p, p') + d(p', q') + d(q', q) < 2\varepsilon + \text{diam } E.$$

Hence $\text{diam } \bar{E} \leq 2\varepsilon + \text{diam } E$ for all $\varepsilon > 0$. $\square$

Proof of 3.10 (b): Intersection Is a Single Point

Step 1 — intersection is nonempty

Each K_n is **nonempty** and **compact**, and $K_n \supset K_{n+1}$, so every finite subfamily has nonempty intersection. By **Theorem 2.36**, $K = \bigcap_{n=1}^{\infty} K_n \neq \emptyset$.



Proof of 3.10 (b): Intersection Is a Single Point

Step 1 — intersection is nonempty

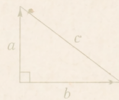
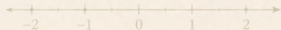
Each K_n is **nonempty** and **compact**, and $K_n \supset K_{n+1}$, so every finite subfamily has nonempty intersection. By **Theorem 2.36**, $K = \bigcap_{n=1}^{\infty} K_n \neq \emptyset$.

Step 2 — it cannot contain two points

If K had two points, $\text{diam } K > 0$. But $K \subset K_n$ for every n , so

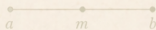
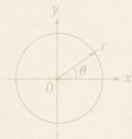
$$\text{diam } K_n \geq \text{diam } K > 0 \quad \text{for all } n,$$

contradicting $\text{diam } K_n \rightarrow 0$. So K is exactly one point. $\square$



$$a^2 + b^2 = c^2$$

The Cauchy Criterion



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

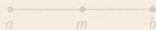
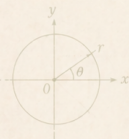
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Theorem 3.11: Cauchy and Convergent

Theorem 3.11 (a)

In any metric space X , every **convergent** sequence is Cauchy.



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

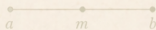
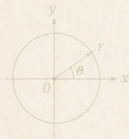
Theorem 3.11: Cauchy and Convergent

Theorem 3.11 (a)

In any metric space X , every **convergent** sequence is Cauchy.

Theorem 3.11 (b)

In a **compact** metric space X , every Cauchy sequence **converges** in X .



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

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Theorem 3.11: Cauchy and Convergent

Theorem 3.11 (a)

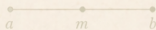
In any metric space X , every **convergent** sequence is Cauchy.

Theorem 3.11 (b)

In a **compact** metric space X , every Cauchy sequence **converges** in X .

Theorem 3.11 (c)

In $\mathbb{R}^k$, every Cauchy sequence **converges**.



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

Why This Theorem Is Powerful

The Cauchy Criterion

The convergence definition involves the limit.
The Cauchy condition does not.

- In $\mathbb{R}^k$: $\{p_n\}$ converges $\iff \{p_n\}$ is Cauchy.
- Decide convergence **without** producing the limit.

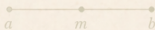
Proof of 3.11 (a): Convergent $\Rightarrow$ Cauchy

Triangle Inequality

Suppose $p_n \rightarrow p$. Given $\varepsilon > 0$, choose N with $d(p_n, p) < \varepsilon$ for $n \geq N$.
Then for $n, m \geq N$,

$$d(p_n, p_m) \leq d(p_n, p) + d(p, p_m) < 2\varepsilon. \quad \square$$

Idea: Triangle inequality routes any two terms through the limit.



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Proof Strategy: 3.11 (b)

Setting: X compact, $\{p_n\}$ Cauchy. Want: $p_n \rightarrow p$ for some $p \in X$.

Plan:

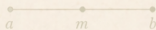
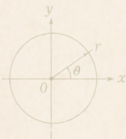
1. Form tails $E_N = \{p_N, p_{N+1}, \dots\}$ and their closures $\bar{E}_N$.
2. Show $\text{diam } \bar{E}_N \rightarrow 0$ (use Theorem 3.10 (a) + Cauchy condition).
3. Each $\bar{E}_N$ is compact and $\bar{E}_N \supset \bar{E}_{N+1}$ (nested compacts).
4. Apply Theorem 3.10 (b): a unique $p \in \bigcap_N \bar{E}_N$.
5. Show $p_n \rightarrow p$.

Proof of 3.11 (b) — Step 1: Tails and Diameters

Define $E_N = \{p_N, p_{N+1}, p_{N+2}, \dots\}$ for each N .

Cauchy $\implies$ shrinking diameters

$$\{p_n\} \text{ Cauchy} \implies \text{diam } E_N \rightarrow 0 \xRightarrow{\text{Thm 3.10 (a)}} \text{diam } \bar{E}_N \rightarrow 0. \quad (3)$$



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

b

Proof of 3.11 (b) — Step 2: Nested Compacts

Closures are nonempty, compact, nested

Each $\bar{E}_N$ contains p_N , hence is **nonempty**.

$\bar{E}_N$ is closed in compact X , hence **compact** (Theorem 2.35).

$E_N \supset E_{N+1}$ gives $\bar{E}_N \supset \bar{E}_{N+1}$.

A unique limit candidate

All hypotheses of Theorem 3.10 (b) hold, so there is a **unique** $p \in X$ with

$p \in \bar{E}_N$ for every N .

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

Proof of 3.11 (b) — Step 3: $p_n \rightarrow p$

Closing the argument

Let $\varepsilon > 0$. By (3), choose N_0 with $\text{diam } \bar{E}_N < \varepsilon$ for $N \geq N_0$.

Since $p \in \bar{E}_N$, every $q \in \bar{E}_N$ satisfies $d(p, q) < \varepsilon$.

In particular, for $n \geq N_0$,

$$d(p, p_n) < \varepsilon, \quad \text{i.e.,} \quad p_n \rightarrow p. \quad \square$$

Proof of 3.11 (c): Cauchy in $\mathbb{R}^k$ Converges

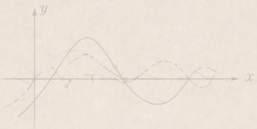
Reduce to part (b)

Let $\{\mathbf{x}_n\}$ be Cauchy in $\mathbb{R}^k$. Some tail E_N has $\text{diam } E_N < 1$, so the range

$$\{\mathbf{x}_n : n \geq 1\} = E_N \cup \{\mathbf{x}_1, \dots, \mathbf{x}_{N-1}\}$$

is **bounded**. By Theorem 2.41, its closure is **compact**.

Apply (b) inside this compact set. $\square$

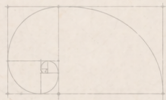


$$e^{i\theta} = \cos \theta + i \sin \theta$$



$$\nabla^2 \phi = 0$$

Complete Metric Spaces



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



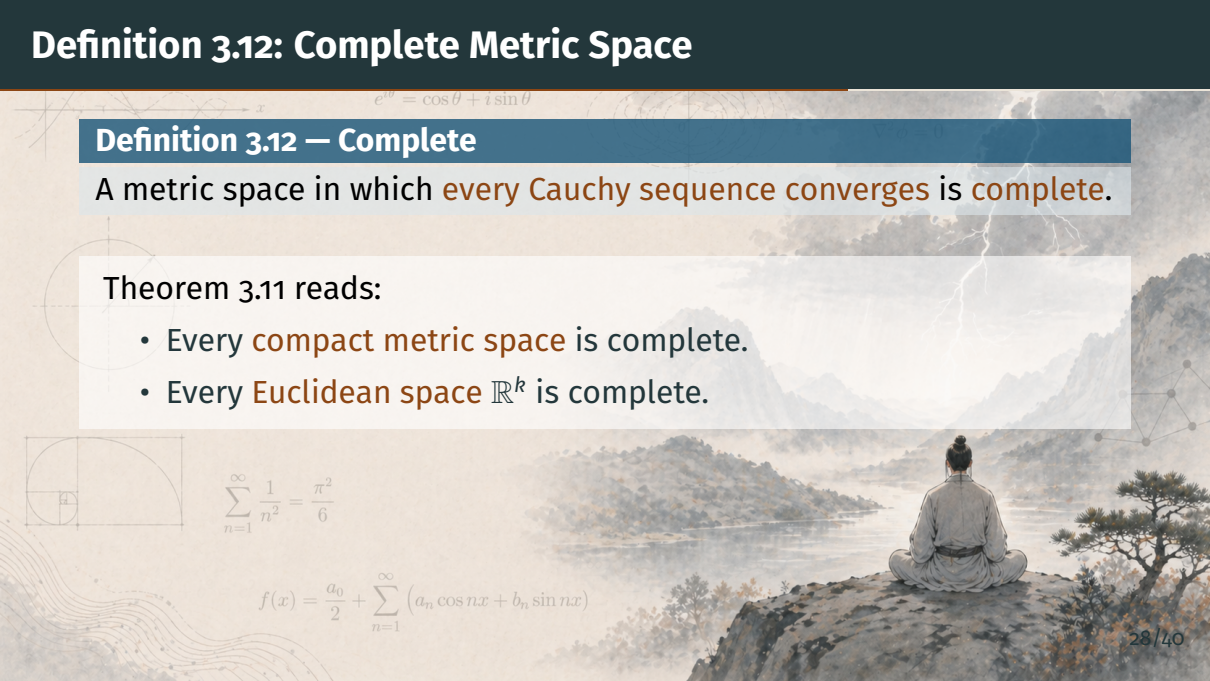
Definition 3.12: Complete Metric Space

Definition 3.12 — Complete

A metric space in which every Cauchy sequence converges is complete.

Theorem 3.11 reads:

- Every compact metric space is complete.
- Every Euclidean space $\mathbb{R}^k$ is complete.



The background features a collage of mathematical and artistic elements. On the left, there is a diagram of a circle with a square inscribed within it, and a smaller square inscribed within the circle. Below this is a diagram of a square with a circle inscribed within it. In the center, there is a formula for the sum of the reciprocals of the squares of the natural numbers. At the bottom, there is a formula for a Fourier series. On the right, there is a traditional Chinese landscape painting of a person sitting on a rock overlooking a river and mountains, with a lightning bolt in the sky. A network diagram is visible on the far right.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Closed Subsets of a Complete Space Are Complete

Inheritance

Every **closed subset** E of a complete metric space X is itself complete.

Why:

- A Cauchy sequence in E is Cauchy in X , so it converges to $p \in X$.
- E is closed $\Rightarrow p \in E$.

Example: $\mathbb{Q}$ Is Not Complete

A Cauchy sequence in $\mathbb{Q}$ with no rational limit

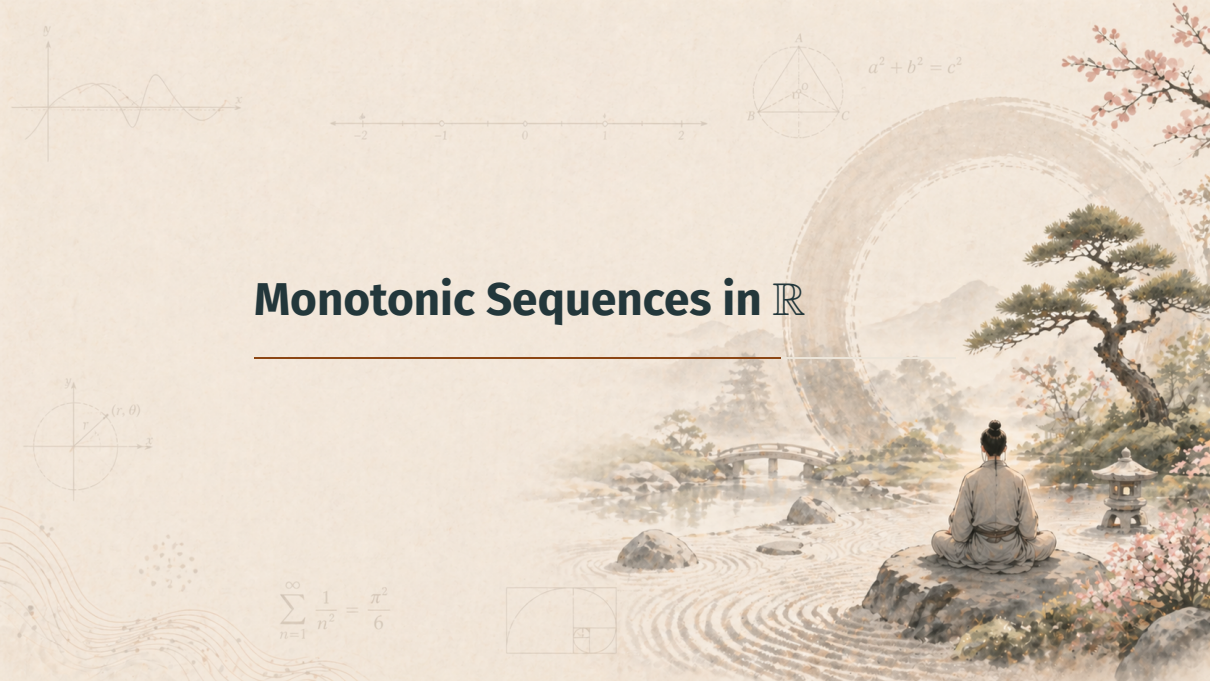
Let r_n be the truncation of $\sqrt{2}$ to n decimal places:

$$r_1 = 1.4, r_2 = 1.41, r_3 = 1.414, r_4 = 1.4142, \dots$$

- Each $r_n \in \mathbb{Q}$ and $|r_n - r_m| \leq 10^{-\min(n,m)} \rightarrow 0$ ($\{r_n\}$ is **Cauchy**).
- In $\mathbb{R}$, $r_n \rightarrow \sqrt{2}$. But $\sqrt{2} \notin \mathbb{Q}$.

So $\{r_n\}$ is Cauchy in $\mathbb{Q}$ **with no limit in $\mathbb{Q}$** .

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



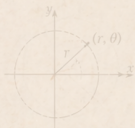
Monotonic Sequences in $\mathbb{R}$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

One Important Case: Monotonic Sequences

Bounded vs. Convergent

- Every convergent sequence in $\mathbb{R}^k$ is bounded (Theorem 3.2 (c)).
- Bounded sequences in $\mathbb{R}^k$ **need not** converge (e.g., $(-1)^n$).
- **Special case:** for monotonic sequences in $\mathbb{R}$, bounded $\iff$ convergent.



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



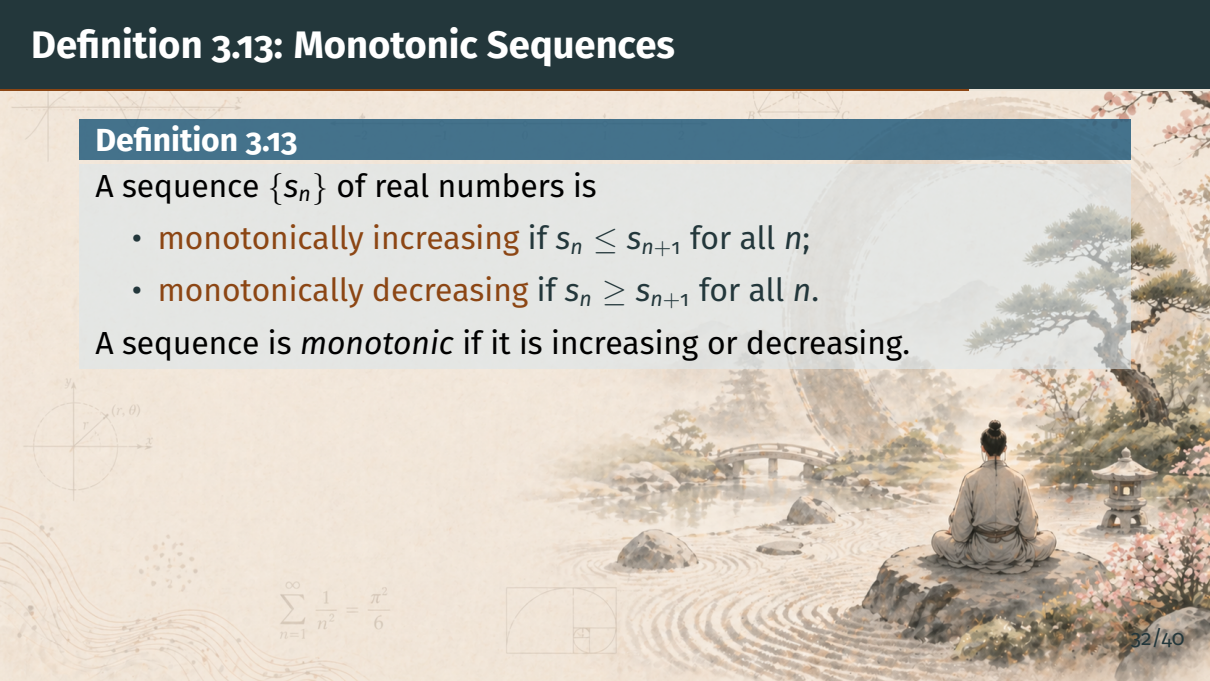
Definition 3.13: Monotonic Sequences

Definition 3.13

A sequence $\{s_n\}$ of real numbers is

- **monotonically increasing** if $s_n \leq s_{n+1}$ for all n ;
- **monotonically decreasing** if $s_n \geq s_{n+1}$ for all n .

A sequence is *monotonic* if it is increasing or decreasing.


$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Theorem 3.14: Bounded $\iff$ Convergent for Monotonic

Theorem 3.14

Suppose $\{s_n\}$ is monotonic. Then

$$\{s_n\} \text{ converges} \iff \{s_n\} \text{ is bounded.}$$

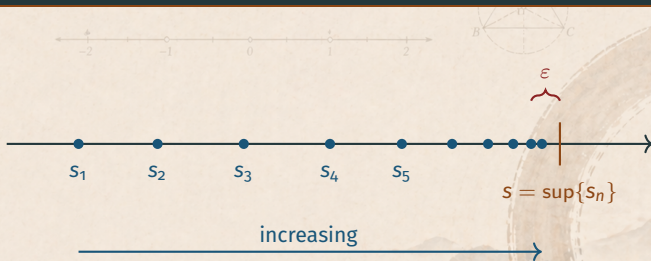
Forward: Theorem 3.2 (c).

Reverse: Use the **least upper bound property** of $\mathbb{R}$.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Picture: Increasing and Bounded $\Rightarrow$ Climbs to the sup



Past index N , s_n enters the strip $(s - \epsilon, s]$ and never leaves.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



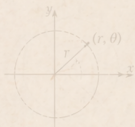
Proof of Theorem 3.14 (Increasing Case)

Setup

Suppose $s_n \leq s_{n+1}$ and $\{s_n\}$ is bounded.

Let $E = \{s_n : n \geq 1\}$ and $s = \sup E$.

Then $s_n \leq s$ for every n .



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Proof of Theorem 3.14 (Increasing Case)

Setup

$$E = \{s_n : n \geq 1\}, \quad s = \sup E, \quad s_n \leq s \text{ for all } n.$$

The supremum is the limit

Fix $\varepsilon > 0$. Since $s - \varepsilon < s$, $s - \varepsilon$ is not an upper bound of E , so

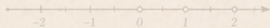
$$\exists N : s - \varepsilon < s_N \leq s.$$

Since $\{s_n\}$ increases, for every $n \geq N$:

$$s - \varepsilon < s_N \leq s_n \leq s. \implies s_n \rightarrow s. \quad \square$$

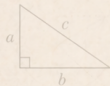
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



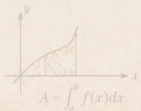


$$x^2 + y^2 = r^2$$

Summary



$$a^2 + b^2 = c^2$$



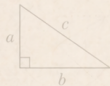
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



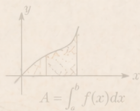
Summary

Key takeaways from this section:

- Definition 3.8: $\{p_n\}$ is **Cauchy** if any two late terms are within ε .



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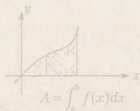
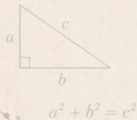


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Summary

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- Definition 3.8: $\{p_n\}$ is **Cauchy** if any two late terms are within ε .
- Theorem 3.11: in compact X (and in $\mathbb{R}^k$), **Cauchy** $\iff$ **convergent**.

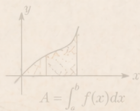
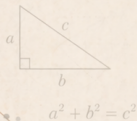


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Summary

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- Definition 3.8: $\{p_n\}$ is **Cauchy** if any two late terms are within ε .
- Theorem 3.11: in compact X (and in $\mathbb{R}^k$), **Cauchy** $\iff$ **convergent**.
- Definition 3.12: a metric space is **complete** if every Cauchy sequence converges.



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Summary

Key takeaways from this section:

- Definition 3.8: $\{p_n\}$ is **Cauchy** if any two late terms are within ε .
- Theorem 3.11: in compact X (and in $\mathbb{R}^k$), **Cauchy** $\iff$ **convergent**.
- Definition 3.12: a metric space is **complete** if every Cauchy sequence converges.
- Theorem 3.14: monotonic sequences in $\mathbb{R}$ **converge** $\iff$ **bounded**.

Coming up next: *Upper and Lower Limits* — naming the two extreme cluster points of any real sequence.